Linear Algebra
(线性代数)

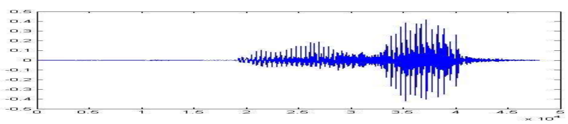
What to learn
in Linear Algebra?

Linear System

System

- A system has input and output (function, transformation, operator)

<u>Speech Recognition System</u>



“How are you”

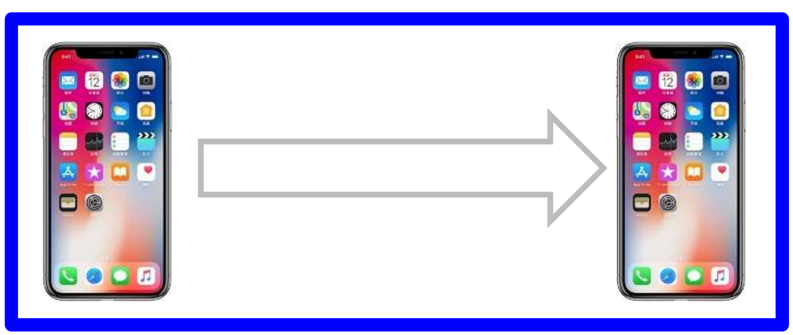
<u>Dialogue System</u> (e.g. Siri, Alexa)

“How are you”

“I am fine”

<u>Communication System</u>

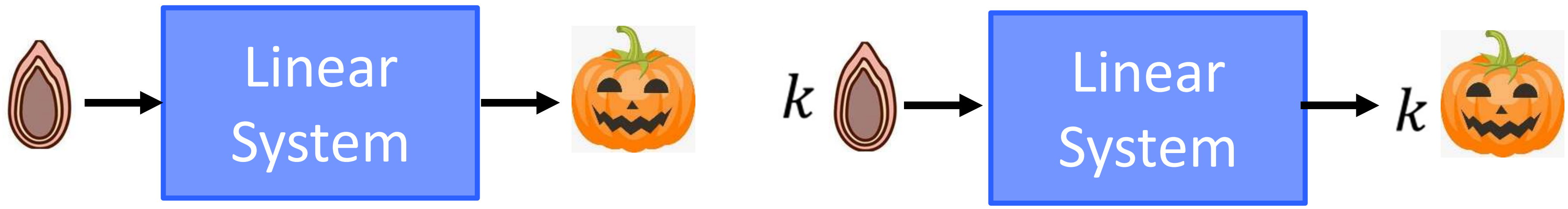
“Hello”



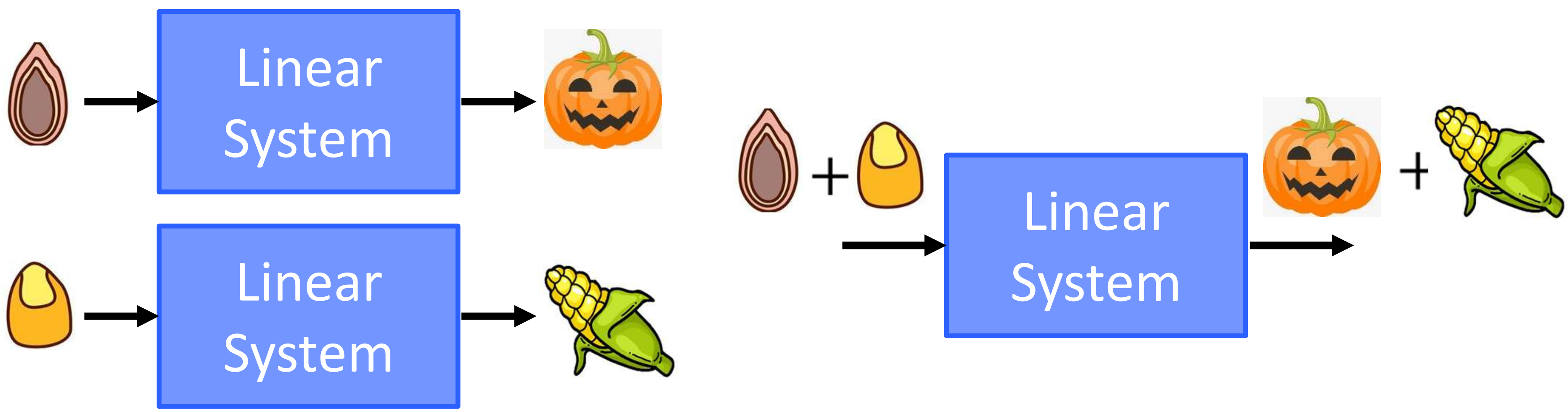
“Hello”

Linear System

- 1. Persevering Multiplication



- 2. Persevering Addition



Linear System

When the input and output are vectors

- 1. Persevering Multiplication

$$\begin{bmatrix}1\\2\end{bmatrix} \rightarrow \boxed{\text{Linear System}} \rightarrow \begin{bmatrix}3\\4\\5\end{bmatrix}$$

$$k\begin{bmatrix}1\\2\end{bmatrix} \rightarrow \boxed{\text{Linear System}} \rightarrow k\begin{bmatrix}3\\4\\5\end{bmatrix}$$

$$k = 2 \quad \begin{bmatrix}2\\4\end{bmatrix} \qquad \begin{bmatrix}6\\8\\10\end{bmatrix}$$

- 2. Persevering Addition

$$\begin{bmatrix}1\\2\end{bmatrix} \rightarrow \boxed{\text{Linear System}} \rightarrow \begin{bmatrix}3\\4\\5\end{bmatrix}$$

$$\begin{bmatrix}5\\6\end{bmatrix} \rightarrow \boxed{\text{Linear System}} \rightarrow \begin{bmatrix}7\\8\\9\end{bmatrix}$$

$$\underset{\begin{bmatrix}6\\8\end{bmatrix}}{\begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}5\\6\end{bmatrix}} \rightarrow \boxed{\text{Linear System}} \rightarrow \underset{\begin{bmatrix}10\\12\\14\end{bmatrix}}{\begin{bmatrix}3\\4\\5\end{bmatrix} + \begin{bmatrix}7\\8\\9\end{bmatrix}}$$

Are they *Linear*?

Linear?

NO

$$x \longrightarrow \boxed{\text{System}} \longrightarrow x^2$$

- 1. Persevering Multiplication

$$x \longrightarrow \boxed{\text{System}} \longrightarrow x^2 \qquad kx \longrightarrow \boxed{\text{System}} \longrightarrow k^2x^2$$

$$\neq kx^2$$

- 2. Persevering Addition

$$x_1 \longrightarrow \boxed{\text{System}} \longrightarrow (x_1)^2$$

$$x_2 \longrightarrow \boxed{\text{System}} \longrightarrow (x_2)^2$$

$$x_1 + x_2 \longrightarrow \boxed{\text{System}} \longrightarrow (x_1 + x_2)^2$$

$$\neq (x_1)^2 + (x_2)^2$$

Linear?

Transpose

YES

$$\begin{bmatrix}6&9\\8&0\\9&2\end{bmatrix} \longrightarrow \boxed{\text{System}} \longrightarrow \begin{bmatrix}6&8&9\\9&0&2\end{bmatrix}$$

$$\Big\downarrow \times k \qquad\qquad \Big\downarrow \times k$$

$$\begin{bmatrix}6k&9k\\8k&0\\9k&2k\end{bmatrix} \longrightarrow \boxed{\text{System}} \longrightarrow \begin{bmatrix}6k&8k&9k\\9k&0&2k\end{bmatrix}$$

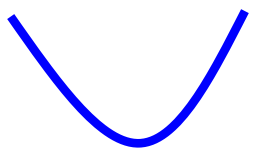
$$\boxed{\begin{bmatrix}6&9\\8&0\\9&2\end{bmatrix}} \longrightarrow \boxed{\text{System}} \longrightarrow \boxed{\begin{bmatrix}6&8&9\\9&0&2\end{bmatrix}}$$

$$\boxed{\begin{bmatrix}1&4\\2&5\\3&6\end{bmatrix}} \longrightarrow \boxed{\text{System}} \longrightarrow \boxed{\begin{bmatrix}1&2&3\\4&5&6\end{bmatrix}}$$

$$\begin{bmatrix}7&13\\10&5\\12&8\end{bmatrix} \longrightarrow \boxed{\text{System}} \longrightarrow \begin{bmatrix}7&10&12\\13&5&8\end{bmatrix}$$

Linear? **YES**

function f → **Derivative** → function f’



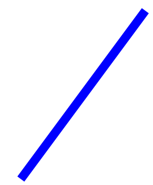
x^2

$$f \to f'$$

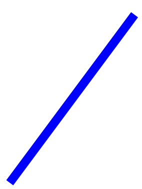
$$g \to g'$$

$$kf \to kf'$$

$$f + g \to f' + g'$$



2x



3x

3

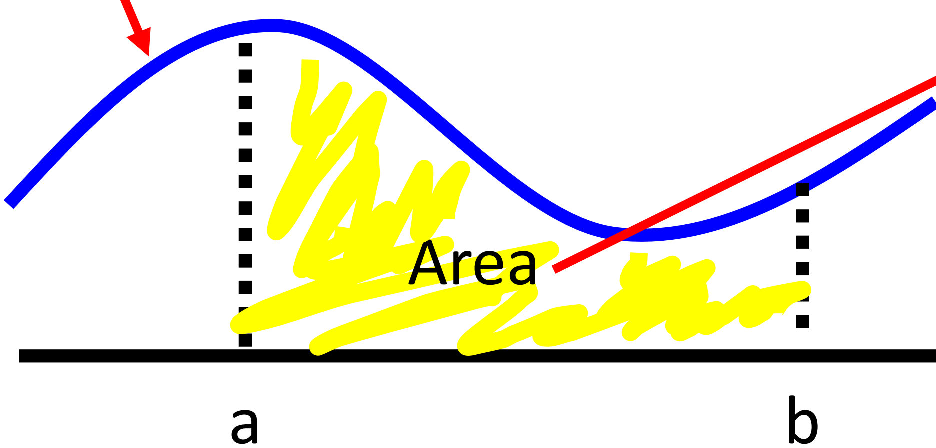
Linear?

$$\frac{1}{3}(b^3 - a^3)$$

$f(x) = x^2$

Function $f(x)$ → **Integral (from a to b)** → $\int_a^b f(x)dx$

scalar



Linear?

YES

Persevering Multiplication

$$f(x) \longrightarrow \boxed{\int_a^b f(x)dx}$$

$$kf(x) \longrightarrow \int_a^b kf(x)dx = k\boxed{\int_a^b f(x)dx}$$

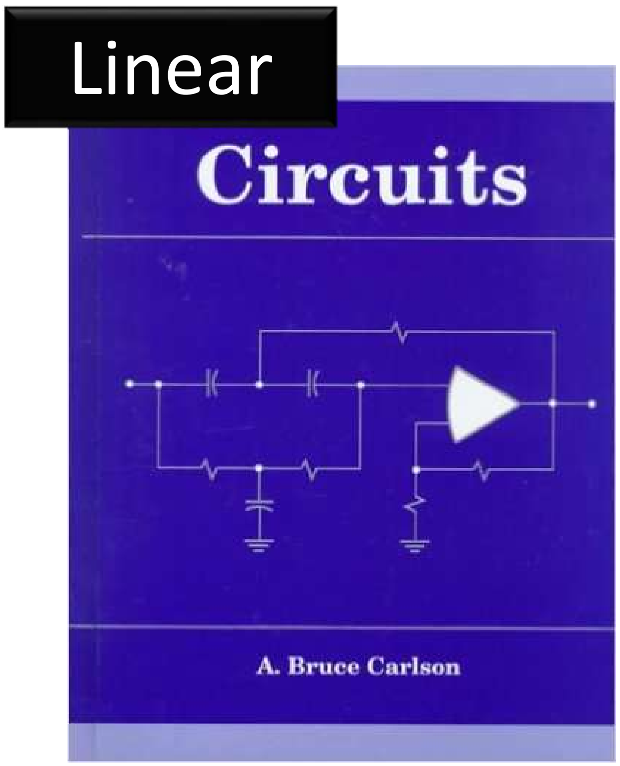
Persevering Addition

$$f(x) \longrightarrow \boxed{\int_a^b f(x)dx} \qquad g(x) \longrightarrow \boxed{\int_a^b g(x)dx}$$

$$f(x) + g(x) \longrightarrow \int_a^b [f(x) + g(x)]dx = \boxed{\int_a^b f(x)dx} + \boxed{\int_a^b g(x)dx}$$

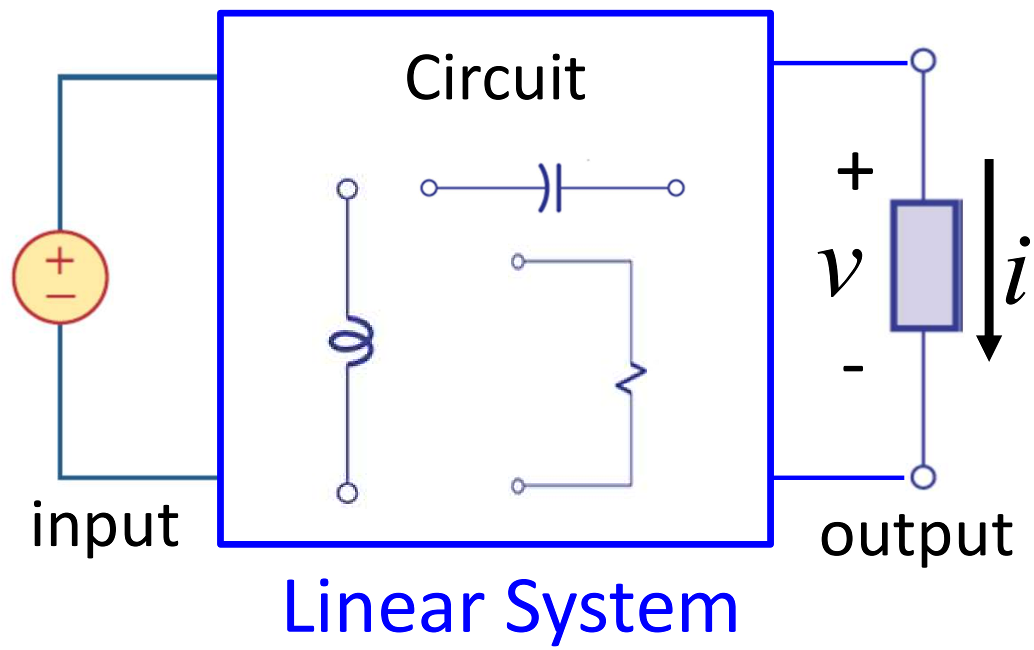
Linear Algebra
v.s.
Other Courses

電路學

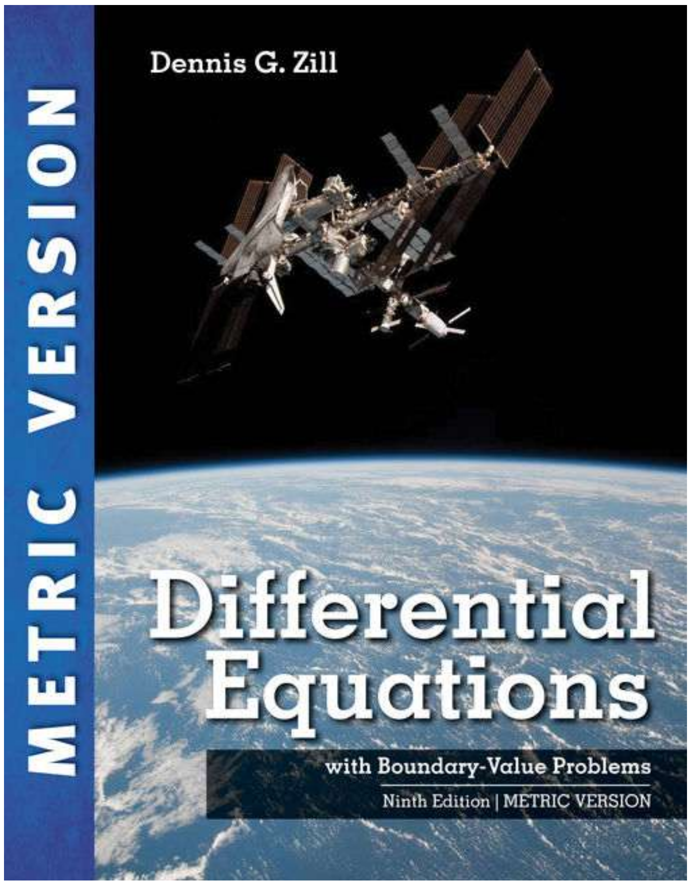


Input: voltage source, current source

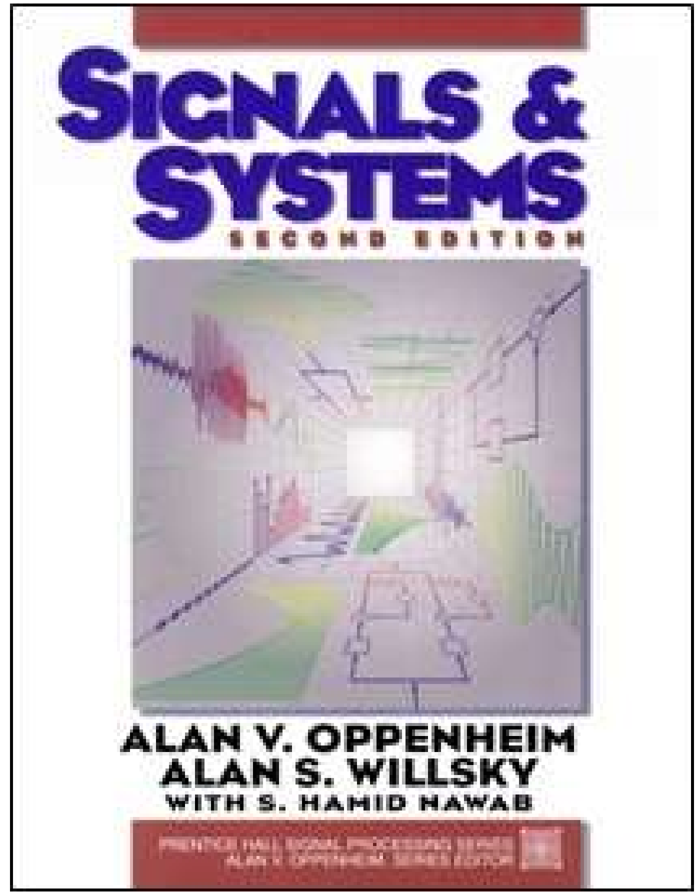
output: voltage and current on the load (燈泡、引擎)



微分方程、信號與系統



METRIC VERSION
Dennis G. Zill
Differential Equations
with Boundary-Value Problems
Ninth Edition | METRIC VERSION



SIGNALS & SYSTEMS
SECOND EDITION
ALAN V. OPPENHEIM
ALAN S. WILLSKY
WITH S. HAMID NAWAB

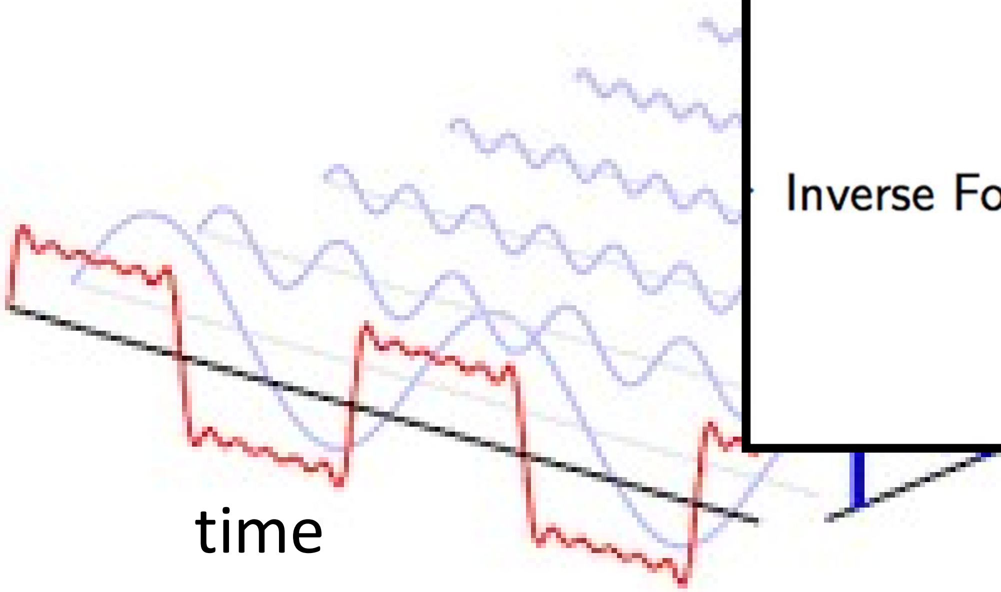
微分方程、信號與系統

Fourier Transform of $x(t)$: $\mathcal{F}[x(t)]$ or $X(w)$:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Inverse Fourier Transform of $X(w)$: $\mathcal{F}^{-1}[X(w)]$:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$



time

frequency

Fourier Transform

Linear System

Complex …

but linear

Brief Course Overview

What to learn?

What is the input? → Linear System → $\begin{bmatrix}1\\2\\3\\4\end{bmatrix}$ Desired output

Chapter 1, Chapter 2

Does it have solution?	Does it have unique solution?	How to find the solution?

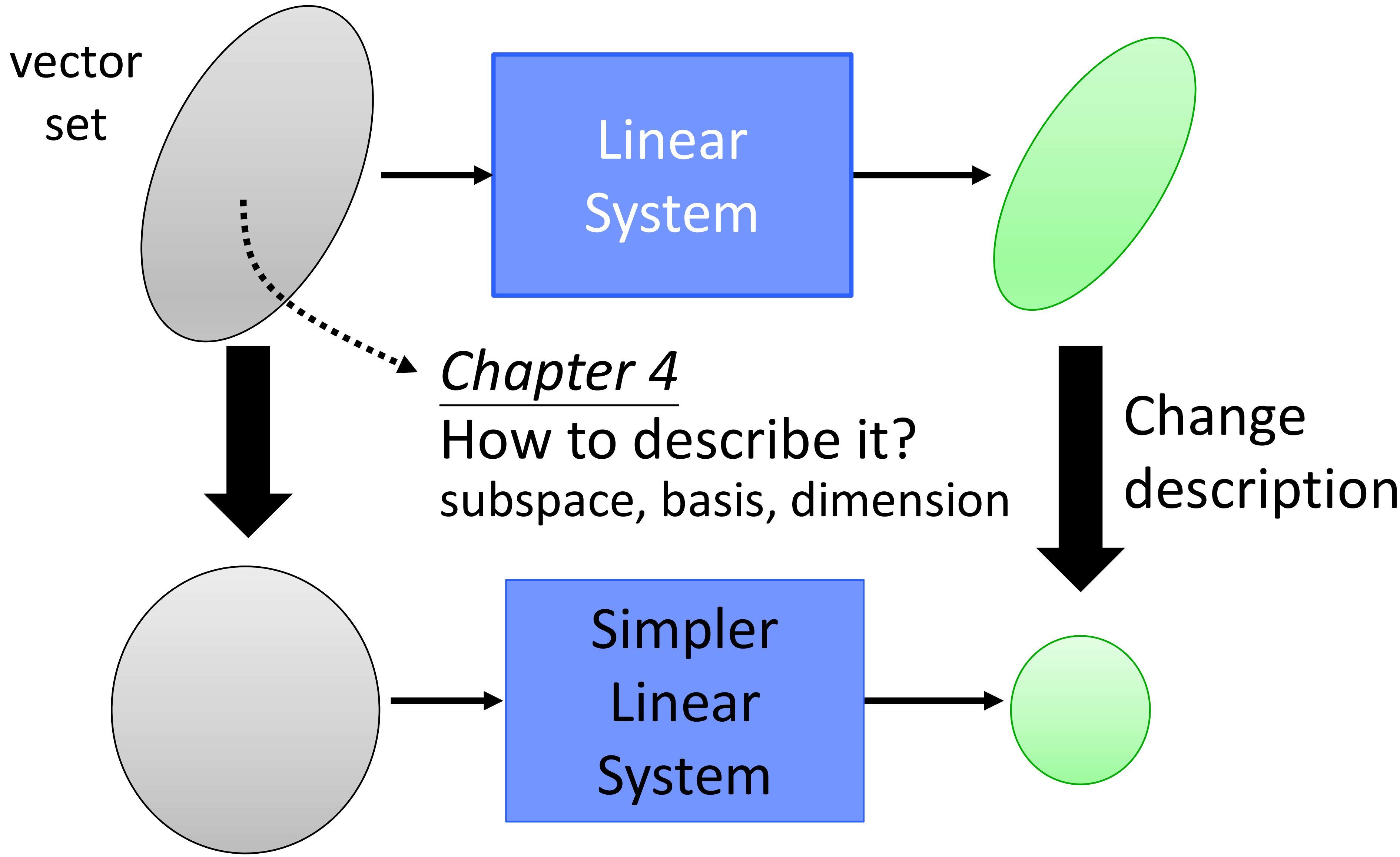
Different views

Chapter 3

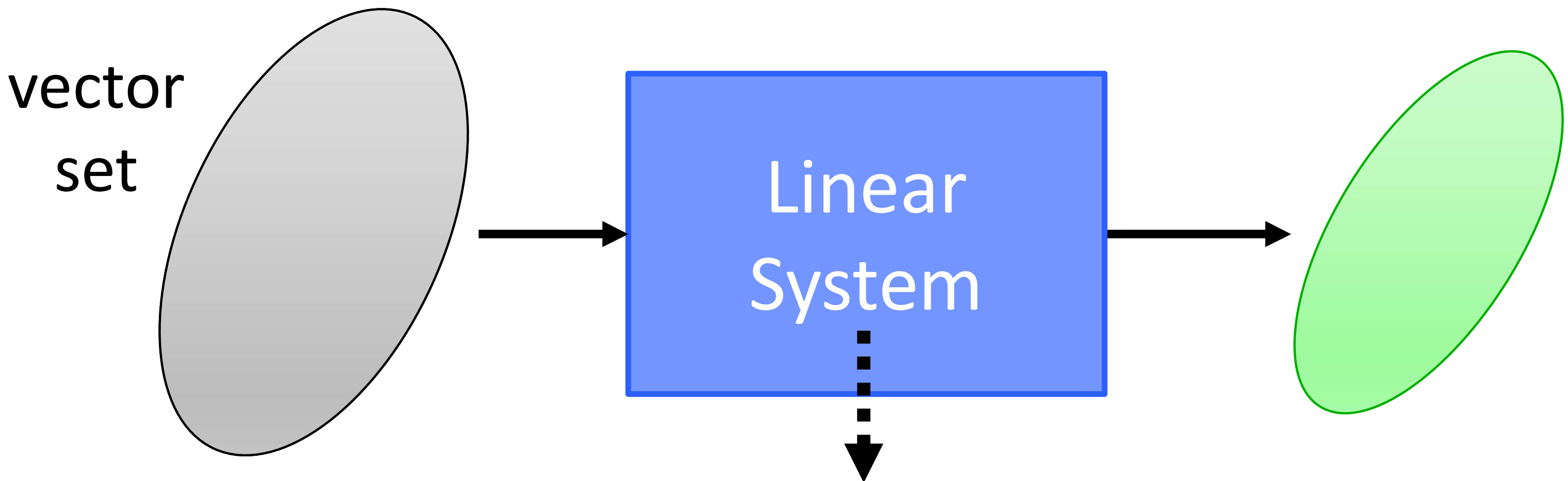
Determinant (行列式)

Beyond 3 X 3

What to learn?



What to learn?



Chapter 5: Eigenvector, Eigenvalue, …


Vectors
THEY ARE EVERYWHERE
meme-arsenal.ru

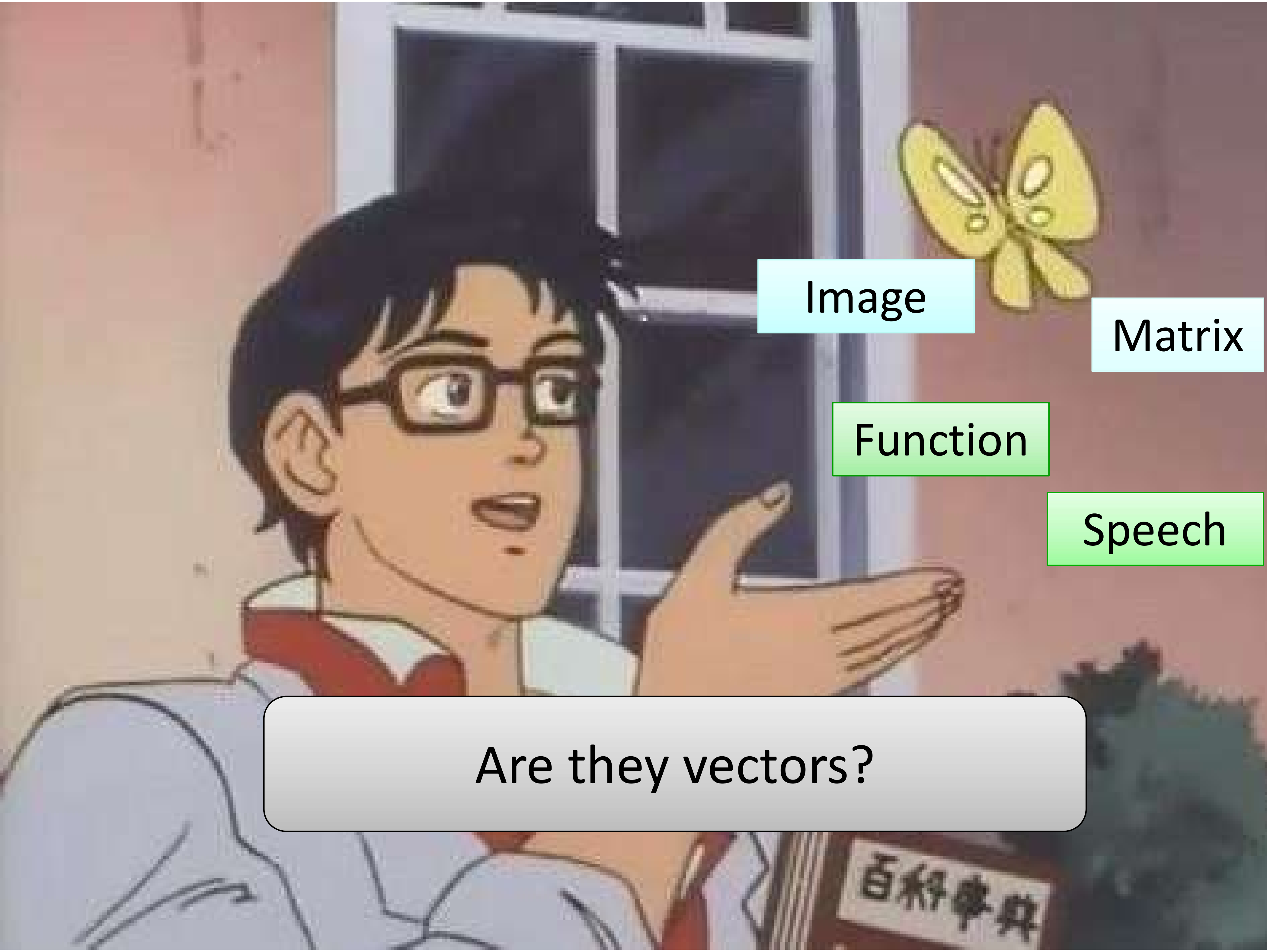


Image
Matrix
Function
Speech
Are they vectors?
百科事典

Chapter 1

Linear Equations in Linear Algebra

Outline

- 1.1 Systems of Linear Equations
- 1.2 Row Reduction and Reduced Echelon Form
- 1.3 Vector Equations
- 1.4 The Matrix Equation
- 1.5 Solution Sets of Linear Systems
- 1.7 Linear Independence

1.1 Systems of Linear Equations
(线性方程组)

Some Terminologies

- **Equation**-----An equation is a mathematical statement that asserts the equality of two expressions, as in x+3=5.
- **Linear Equation------**

A linear equation in the variables $x_1, x_2, \ldots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where b and the coefficients $a_1, a_2, \ldots, a_n$ are real numbers, usually known in advance. The subscript n may be any positive integer.

Character of linear equations: each term is either a constant or the product of a constant and (the first power of) a single variable.

The equations $4x_1 - 5x_2 + 2 = x_1$ are linear because it can be rearranged as $3x_1 - 5x_2 = -2$;

The equation $4x_1 - 5x_2 = x_1 x_2$ is not linear because of the presence of $x_1 x_2$.

- **System of Linear Equations** (or **Linear System**)---------a collection of one or more linear equations involving the same variables, say, $x_1, x_2, \cdots, x_n$.

An example is $\begin{cases} 2x_1 - 5x_2 + x_3 = 6 \\ 3x_1 \qquad\qquad +4x_3 = 2 \end{cases}$.

Linear equation in n unknowns/variables:

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

$$\Rightarrow a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$$

System of m linear equations in n unknowns:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots & \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & b_m \end{cases}$$

Linear equation in n unknowns/variables:

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

$$\Rightarrow a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$$

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Linear equation in n unknowns/variables:

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

$$\Rightarrow a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$$

System of m linear equations in n unknowns:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots & \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & b_m \end{cases}$$

Call it $m \times n$ (linear) system.

$$\text{(a)}\begin{cases} x_1 + 2x_2 = 5 \\ 2x_1 + 3x_2 = 8 \end{cases} \qquad 2\times 2 \qquad (1,2)$$

$$\text{(b)}\begin{cases} x_1 - x_2 + x_3 = 2 \\ 2x_1 + x_2 - x_3 = 4 \end{cases} \qquad 2\times 3 \qquad (2,a,a)\ (a\in R)$$

$$\text{(c)}\begin{cases} x_1 + x_2 = 2 \\ x_1 - x_2 = 1 \\ x_1 = 4 \end{cases} \qquad 3\times 2 \qquad \text{no solution}$$

A solution of an $m\times n$ system

---an ordered n-tuple of numbers $(x_1, x_2, \ldots, x_n)$ that satisfies all the equations of the system.

A system of linear equations is said to be **consistent(相容的)** if it has either one solution or infinitely many solutions;

A system is **inconsistant(不相容的)** if it has no solution.

Solution set (解集)

= the collection of solution(s) of a system

Solve a consistent linear system ⟺ Find its solution set

Question

- What are we going to study?

Let us illustrate with simple examples:

(a)
$$\begin{aligned} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 2 \end{aligned}$$

(b)
$$\begin{aligned} x_1 + x_2 &= 2 \\ x_1 + x_2 &= 1 \end{aligned}$$

(c)
$$\begin{aligned} x_1 + x_2 &= 2 \\ -x_1 - x_2 &= -2 \end{aligned}$$

Problems:

- What are the possible types of solutions?

Question

- What are we going to study?

Let us illustrate with simple examples:

(a) $\begin{aligned} x_1 + x_2 &= \ 2 \\ x_1 - x_2 &= \ 2 \end{aligned}$ (b) $\begin{aligned} x_1 + x_2 &= \ 2 \\ x_1 + x_2 &= \ 1 \end{aligned}$ (c) $\begin{aligned} x_1 + x_2 &= \ 2 \\ -x_1 - x_2 &= \ -2 \end{aligned}$

Problems:

- What are the possible types of solutions?
- How to determine the solution type?

Question

- What are we going to study?

Let us illustrate with simple examples:

(a) $\begin{aligned} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 2 \end{aligned}$ (b) $\begin{aligned} x_1 + x_2 &= 2 \\ x_1 + x_2 &= 1 \end{aligned}$ (c) $\begin{aligned} x_1 + x_2 &= 2 \\ -x_1 - x_2 &= -2 \end{aligned}$

Problems:

- What are the possible types of solutions?
- How to determine the solution type?
- How to see if two systems represent the same solutions?

Question

- What are we going to study?

Let us illustrate with simple examples:

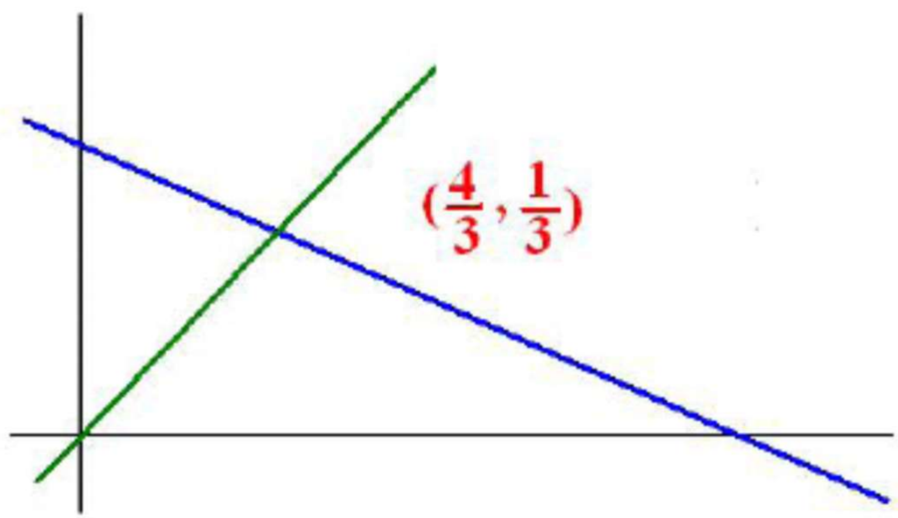
$$\text{(a)}\quad \begin{aligned} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 2 \end{aligned} \qquad \text{(b)}\quad \begin{aligned} x_1 + x_2 &= 2 \\ x_1 + x_2 &= 1 \end{aligned} \qquad \text{(c)}\quad \begin{aligned} x_1 + x_2 &= 2 \\ -x_1 - x_2 &= -2 \end{aligned}$$

Problems:

- What are the possible types of solutions?
- How to determine the solution type?
- How to see if two systems represent the same solutions?
- How to solve a "big" system?

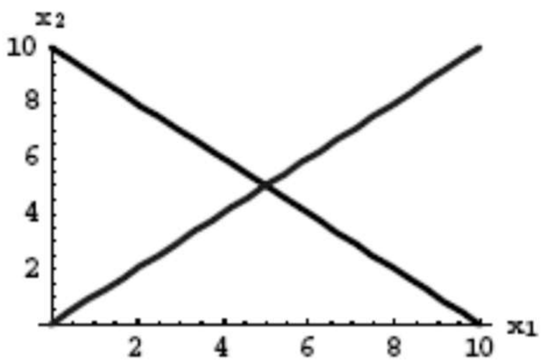
➢Geometric Interpretation of systems of linear equations

Example. What does the solution of $\begin{cases} x+2y = 2 \\ x-y = 1 \end{cases}$ mean?

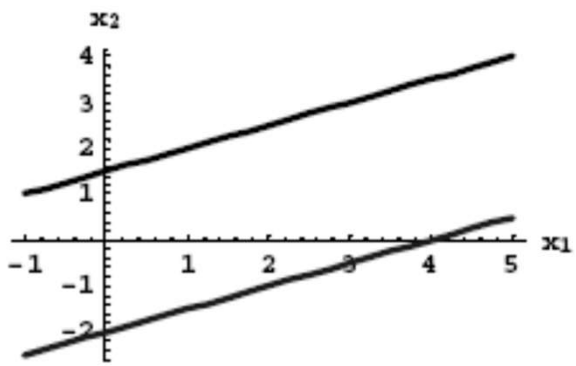


EXAMPLE Two equations in two variables:

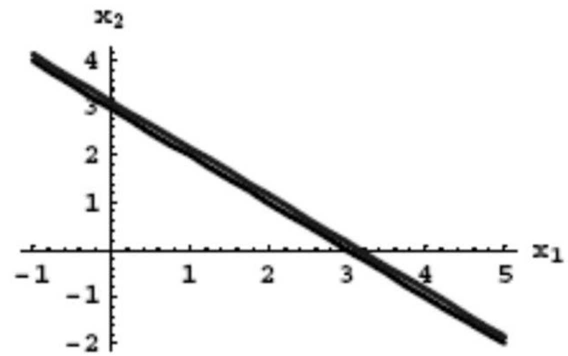
$$\begin{aligned} x_1 + x_2 &= 10 \\ -x_1 + x_2 &= 0 \end{aligned} \qquad \begin{aligned} x_1 - 2x_2 &= -3 \\ 2x_1 - 4x_2 &= 8 \end{aligned} \qquad \begin{aligned} x_1 + x_2 &= 3 \\ -2x_1 - 2x_2 &= -6 \end{aligned}$$



one unique solution



no solution



infinitely many solutions

We use the bracket $\{\ \cdots\ \}$ to collect the solutions.

Reconsider the example:

Example.

(a) $$\begin{aligned} x_1 + x_2 &= 2 \\ x_1 - x_2 &= 2 \end{aligned}$$

The solution set is $\{(2,0)\}$.

(b) $$\begin{aligned} x_1 + x_2 &= 2 \\ x_1 + x_2 &= 1 \end{aligned}$$

The solution set is $\{\quad\}$, denoted by $\emptyset$.

(c) $x_1 + x_2 = \quad 2$

$$-x_1 - x_2 = \quad -2$$

The solution set is $\{(a, 2-a):\ a$ is a real number$\}$.

We can describe a solution set in different ways:

$\{(2-\alpha, \alpha):\ \alpha$ is a real number$\}$,

$\{(1-t, 1+t):\ t$ is a real number$\}$.

BASIC FACT: A system of linear equations has either
(i) exactly one solution *(consistent)* or
(ii) infinitely many solutions *(consistent)* or
(iii) no solution *(inconsistent)*.

Equivalent Systems (等价系统)

Two systems involving the same variables are equivalent if they have the same solution sets.

That is, each solution of the first system is a solution of the second system, and each solution of the second system is a solution of the first.

Example. the following systems are equivalent.

(a)
$$\begin{aligned} x_1 + x_2 &= 2 \\ -x_1 - x_2 &= -2 \end{aligned}$$

(b) $x_1 + x_2 = 2$

Example. the following systems are equivalent.

(i)
$$\begin{aligned} 3x_1 + 2x_2 - x_3 &= -2 \\ x_2 \qquad &= 3 \\ 2x_3 &= 4 \end{aligned}$$

(ii)
$$\begin{aligned} 3x_1 + 2x_2 - x_3 &= -2 \\ -3x_1 - x_2 + x_3 &= 5 \\ 3x_1 + 2x_2 + x_3 &= 2 \end{aligned}$$

- A system is characterized by its solution set.
- How to solve a system? Elimination(消元)

Example. Solve $\begin{cases} 2x-y = 9 \\ -x+7y = -10 \end{cases}$

Solution.

$$\begin{cases} 2x-y = 9 \\ -x+7y = -10 \end{cases} \longrightarrow \begin{cases} -x+7y = -10 \\ 2x-y = 9 \end{cases}$$

Interchange the equations

- A system is characterized by its solution set.
- How to solve a system?

Example. Solve $\begin{cases} 2x-y = 9 \\ -x+7y = -10 \end{cases}$

Solution.

$$\begin{cases} 2x-y = 9 \\ -x+7y = -10 \end{cases} \longrightarrow \begin{cases} -x+7y = -10 \\ 2x-y = 9 \end{cases}$$

$$\begin{cases} -2x+14y = -20 \\ 2x-y = 9 \end{cases}$$

Multiply 2 to 1st equation

- A system is characterized by its solution set.
- How to solve a system?

Example. Solve $\begin{cases} 2x-y &= 9 \\ -x+7y &= -10 \end{cases}$

Solution.

$$\begin{cases} 2x-y &= 9 \\ -x+7y &= -10 \end{cases} \longrightarrow \begin{cases} -x+7y &= -10 \\ 2x-y &= 9 \end{cases}$$

$$\begin{cases} -2x+14y &= -20 \\ 2x-y &= 9 \end{cases} \longrightarrow \begin{cases} -2x+14y &= -20 \\ 13y &= -11 \end{cases}$$

Add 1st equation to the 2nd

- A system is characterized by its solution set.
- How to solve a system?

Equivalent system

Example. Solve $\begin{cases} 2x - y = 9 \\ -x + 7y = -10 \end{cases}$

Solution.

Echelon form

$$\begin{cases} 2x - y = 9 \\ -x + 7y = -10 \end{cases} \longrightarrow \begin{cases} -x + 7y = -10 \\ 2x - y = 9 \end{cases}$$

$$\begin{cases} -2x + 14y = -20 \\ 2x - y = 9 \end{cases} \longrightarrow \begin{cases} -2x + 14y = -20 \\ 13y = -11 \end{cases}$$

Solve for y, x by backward substitution

Back substitution

*Step*1. The nth equation is solved for the value of x_n;

*Step*2. The value x_n is used in the $(n-1)$st equation to solve for x_{n-1};

*Step*3. The values x_n and x_{n-1} are used in the $(n-2)$nd equation to solve for x_{n-2};

$\vdots$

✓We can use the ***back substitution*** to solve any echelon form system.

To summarize, there are three operations that can be used on a system to obtain an equivalent system:

I. The order in which any two equations are written may be interchanged. (Interchange)

II. Both sides of an equation may be multiplied by the same nonzero real number. (Scaling)

III. A multiple of one equation may be added to (or subtracted from) another. (Replacement)

✓Given a system of equations, we may use these operations to obtain an equivalent system that is easier to solve.

For a $m \times n$ system

$$\left\{\begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots & \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & b_m \end{array}\right.$$

Observation

The variables and equality signs do not take part.

For a $m\times n$ system

$$\begin{cases} a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n &= b_1 \\ a_{21}x_1+a_{22}x_2+\cdots+a_{2n}x_n &= b_2 \\ \quad\vdots & \vdots \quad \vdots \\ a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n &= b_m \end{cases}$$

we associate with this system a $m\times n$ array of numbers whose entries are the coefficients of the x_i's:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Coefficient matrix
（系数矩阵）

Some terminologies:

1. The term *matrix* (矩阵) means simply a rectangular array of numbers.
2. A matrix having m rows and n columns is said to be $m \times n$.

For a $m \times n$ system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots & \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & b_m \end{cases}$$

If we attach to the coefficient matrix an additional column whose entries are the numbers on the right-hand side of the system, we obtain the new matrix

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & \\ a_{21} & a_{22} & \cdots & a_{2n} & \\ \vdots & \vdots & \vdots & \vdots & \\ a_{m1} & a_{m2} & \cdots & a_{mn} & \end{pmatrix}$$

For a $m \times n$ system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{cases}$$

If we attach to the coefficient matrix an additional column whose entries are the numbers on the right-hand side of the system, we obtain the new matrix

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right)$$

Augmented matrix
(增广矩阵）

The vertical line - separate coefficients & constant terms

Example. Write down the augmented matrix of

$$\begin{cases} x_1 + 2x_2 + x_3 &= 3 \\ 3x_1 - x_2 - 3x_3 &= -1 \\ 2x_1 + 3x_2 + x_3 &= 4 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 3 & -1 & -3 & -1 \\ 2 & 3 & 1 & 4 \end{array}\right)$$

Note: In general, when an $m \times r$ matrix B is attached to an $m \times n$ matrix A in this way, the augmented matrix is denoted by $(A \mid B)$, or $(A \;\; B)$, which size is $m \times (n+r)$.

Objective ----

use matrix language to express the elimination procedure.

The whole course---all the key ideas get expressed as matrix operations, not as words.

Example. Solve $\begin{cases} 2x - y &= 9 \\ -x + 7y &= -10 \end{cases}$

Solution. $\begin{cases} 2x - y &= 9 \\ -x + 7y &= -10 \end{cases} \longrightarrow \begin{cases} -x + 7y &= -10 \\ 2x - y &= 9 \end{cases}$

$$\begin{cases} -2x + 14y &= -20 \\ 2x - y &= 9 \end{cases} \longrightarrow \begin{cases} -2x + 14y &= -20 \\ 13y &= -11 \end{cases}$$

Example. Solve $\begin{cases} 2x - y &= 9 \\ -x + 7y &= -10 \end{cases}$

Solution. $\begin{cases} 2x - y &= 9 \\ -x + 7y &= -10 \end{cases} \longrightarrow \begin{cases} -x + 7y &= -10 \\ 2x - y &= 9 \end{cases}$

$$\begin{cases} -2x + 14y &= -20 \\ 2x - y &= 9 \end{cases} \longrightarrow \begin{cases} -2x + 14y &= -20 \\ 13y &= -11 \end{cases}$$

In terms of augmented matrix,

$$\left(\begin{array}{cc|c} 2 & -1 & 9 \\ -1 & 7 & -10 \end{array}\right)$$

Example. Solve $\begin{cases} 2x - y = 9 \\ -x + 7y = -10 \end{cases}$

Interchange the equations

Solution. $\begin{cases} 2x - y = 9 \\ -x + 7y = -10 \end{cases} \longrightarrow \begin{cases} -x + 7y = -10 \\ 2x - y = 9 \end{cases}$

$\begin{cases} -2x + 14y = -20 \\ 2x - y = 9 \end{cases} \longrightarrow \begin{cases} -2x + 14y = -20 \\ 13y = -11 \end{cases}$

In terms of augmented matrix,

Interchange rows

$$\left(\begin{array}{cc|c} 2 & -1 & 9 \\ -1 & 7 & -10 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} -1 & 7 & -10 \\ 2 & -1 & 9 \end{array}\right)$$

Example. Solve $\begin{cases} 2x - y &= 9 \\ -x + 7y &= -10 \end{cases}$

Multiply 2 to 1st equation

Solution. $\begin{cases} 2x - y &= 9 \\ -x + 7y &= -10 \end{cases} \longrightarrow \begin{cases} -x + 7y &= -10 \\ 2x - y &= 9 \end{cases}$

$$\begin{cases} -2x + 14y &= -20 \\ 2x - y &= 9 \end{cases} \longrightarrow \begin{cases} -2x + 14y &= -20 \\ 13y &= -11 \end{cases}$$

Multiply 2 to 1st row

In terms of augmented matrix,

$$\left(\begin{array}{cc|c} 2 & -1 & 9 \\ -1 & 7 & -10 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} -1 & 7 & -10 \\ 2 & -1 & 9 \end{array}\right) \longrightarrow$$

$$\left(\begin{array}{cc|c} -2 & 14 & -20 \\ 2 & -1 & 9 \end{array}\right)$$

Example. Solve $\begin{cases} 2x - y = 9 \\ -x + 7y = -10 \end{cases}$

Add 1st equation to the 2nd

Solution. $\begin{cases} 2x - y = 9 \\ -x + 7y = -10 \end{cases} \longrightarrow \begin{cases} -x + 7y = -10 \\ 2x - y = 9 \end{cases}$

$$\begin{cases} -2x + 14y = -20 \\ 2x - y = 9 \end{cases} \longrightarrow \begin{cases} -2x + 14y = -20 \\ 13y = -11 \end{cases}$$

Add 1st row to the 2nd

In terms of augmented matrix,

$$\left(\begin{array}{cc|c} 2 & -1 & 9 \\ -1 & 7 & -10 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} -1 & 7 & -10 \\ 2 & -1 & 9 \end{array}\right) \longrightarrow$$

$$\left(\begin{array}{cc|c} -2 & 14 & -20 \\ 2 & -1 & 9 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} -2 & 14 & -20 \\ 0 & 13 & -11 \end{array}\right)$$

- The above operations are sufficient to solve any system of linear equations!

Elementary Row Operations (初等行变换)

I. Interchange two rows (Interchange)

II. Multiply a row by a nonzero real number (Scaling)

III. Replace a row by its sum with a multiple of another row (Replacement)

Notation

I. $R_i \longleftrightarrow R_j$ (interchange ith & jth rows)

II. αR_i (Multiply ith row by α)

III. $\alpha R_i + R_j$ (Replace jth row by $\alpha R_i + R_j$) or (Add αR_i to R_j)

Example. Solve $\begin{cases} x_1 + 2x_2 + x_3 &= 3 \\ 3x_1 - x_2 - 3x_3 &= -1 \\ 2x_1 + 3x_2 + x_3 &= 4 \end{cases}$

Solution. The augmented matrix is $\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 3 & -1 & -3 & -1 \\ 2 & 3 & 1 & 4 \end{array}\right)$

Apply the operations: $-3R_1 + R_2, -2R_1 + R_3, -\frac{1}{7}R_2 + R_3,$

the augmented matrix becomes

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -7 & -6 & -10 \\ 0 & 0 & -\frac{1}{7} & -\frac{4}{7} \end{array}\right)$$

We can read the solution from it- write it back into equations.

Basic idea of solving any linear system

Elementary Row Reduction Algorithm

----the methodology of reducing matrix to an equivalent simper form

- Why is this methodology feasible?

Row equivalent matrices: Two matrices where one matrix can be transformed into the other matrix by a sequence of elementary row operations.

Fact about Row Equivalence: If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

- When will this process stop?

Until we can read the solution from the new matrix.

- How to read the solution?

- write it back into equations.

Simper form:

Echelon form （阶梯形） ➡ Gaussian Elimination algorithm

⇩

Reduced echelon form ➡ Gaussian-Jordan Elimination algorithm

（行简化阶梯形）

Some terminologies in elementary row reduction algorithm:

- a nonzero row or column in a matrix
  ----a row or column that contains at least one nonzero entry;

- a pivot column in a matrix (转轴列)
  ----the leftmost nonzero column;

- a pivot (转轴元)
  ----the **nonzero** entry which is placed at the top of the pivot column;

- a pivot row in a matrix (转轴行)
  ----the row that contains a pivot.

Elementary Row Reduction Algorithm

Step 1.

Begin with the leftmost nonzero column. This is a pivot column.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 3 & -1 & -3 & -1 \\ 2 & 3 & 1 & 4 \end{array}\right)$$

Elementary Row Reduction Algorithm

Step 2.

Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the top of the pivot column. And pick a row with pivot entry to be the pivot row.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 3 & -1 & -3 & -1 \\ 2 & 3 & 1 & 4 \end{array}\right)$$

Elementary Row Reduction Algorithm

Step 3.

Perform elementary row operations to eliminate other entries in the pivot column. (that is, to create zeros in all positions below the pivot.)

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 3 & -1 & -3 & -1 \\ 2 & 3 & 1 & 4 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -7 & -6 & -10 \\ 0 & -1 & -1 & -2 \end{array}\right)$$

Elementary Row Reduction Algorithm

Step 4.

Apply steps 1-3 to the remaining smaller submatrix.

Repeat the process until there are no more nonzero rows to modify.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 3 & -1 & -3 & -1 \\ 2 & 3 & 1 & 4 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -7 & -6 & -10 \\ 0 & -1 & -1 & -2 \end{array}\right)$$

$$\longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -7 & -6 & -10 \\ 0 & 0 & -\frac{1}{7} & -\frac{4}{7} \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 0 & -1 & -1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 6 \\ 2 & 4 & 1 & -2 & -1 \\ 3 & 1 & -2 & 2 & 3 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 0 & -1 & -1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 6 \\ 2 & 4 & 1 & -2 & -1 \\ 3 & 1 & -2 & 2 & 3 \end{array}\right)$$

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Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 2 & 4 & 1 & -2 & -1 \\ 3 & 1 & -2 & 2 & 3 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{rrrr|r} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 2 & -1 & -4 & -13 \\ 0 & -2 & -5 & -1 & -15 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 2 & -1 & -4 & -13 \\ 0 & -2 & -5 & -1 & -15 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 2 & -1 & 4 & -13 \\ 0 & -2 & -5 & -1 & -15 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & -3 & -2 & -13 \\ 0 & 0 & -3 & -3 & -15 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & -3 & -2 & -13 \\ 0 & 0 & -3 & -3 & -15 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & -3 & -2 & -13 \\ 0 & 0 & -3 & -3 & -15 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & -3 & -2 & -13 \\ 0 & 0 & 0 & -1 & -2 \end{array}\right)$$

Example. Solve

$$\begin{cases} -x_2 - x_3 + x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 6 \\ 2x_1 + 4x_2 + x_3 - 2x_4 &= -1 \\ 3x_1 + x_2 - 2x_3 + 2x_4 &= 3 \end{cases}$$

Solution. The augmented matrix is

Echelon form

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & -3 & -2 & -13 \\ 0 & 0 & 0 & -1 & -2 \end{array}\right)$$

The solution set is $\{(2, -1, 3, 2)\}$.

- Augmented matrix in **Echelon form**-tell solution immediately

Row Echelon Form （阶梯形）

(i) The number of leading zeros of nonzero rows strictly increases from the top to the bottom.

Leftmost

(ii) All zero rows, if any, are at the bottom.

- a leading entry of a nonzero row

----the leftmost nonzero entry from the left in a nonzero row, that is, the first nonzero entry in a nonzero row.

EXAMPLE: Echelon forms

(a) $\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} \blacksquare & * & * \\ 0 & \blacksquare & * \\ 0 & 0 & \blacksquare \\ 0 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * \end{bmatrix}$

Examples:

$$\begin{pmatrix} 2 & 4 & 6 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Non-examples:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

How to get the echelon form by the elementary row reduction algorithm?

----Perform the elementary row reduction algorithm.

Example. Row reduce matrix $A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 2 & -1 & 2 & 4 \\ 1 & -1 & 0 & 3 \\ 4 & 1 & 4 & 2 \end{bmatrix}$ to echelon form.

$$\text{Sol. } A \to \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & -3 & -2 & 2 \\ 0 & -2 & -2 & 2 \\ 0 & -3 & -4 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -3 & -2 & 2 \\ 0 & -3 & -4 & -2 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & -5 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Exercise. Row reduce matrix $A=\begin{bmatrix} 1 & -2 & 2 & -1 & 1 \\ 2 & -4 & 8 & 0 & 2 \\ -2 & 4 & -2 & 3 & 3 \\ 3 & -6 & 0 & -6 & 4 \end{bmatrix}$ to echelon form.

$$\text{Sol}.\ A\to\begin{bmatrix} 1 & -2 & 2 & -1 & 1 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 2 & 1 & 5 \\ 0 & 0 & -6 & -3 & 1 \end{bmatrix}\to\begin{bmatrix} 1 & -2 & 2 & -1 & 1 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 & 5 \\ 0 & 0 & -6 & -3 & 1 \end{bmatrix}\to\begin{bmatrix} 1 & -2 & 2 & -1 & 1 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\to\begin{bmatrix} 1 & -2 & 2 & -1 & 1 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Example. Solve the system

$$\left\{\begin{array}{rcl} x_1+x_2+x_3+x_4+x_5 & = & 1 \\ -x_1-x_2+x_5 & = & -1 \\ -2x_1-2x_2+3x_5 & = & 1 \\ x_3+x_4+3x_5 & = & 3 \\ x_1+x_2+2x_3+2x_4+4x_5 & = & 4 \end{array}\right.$$

if its augmented matrix is reduced to

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The solution set is $\{(4-t,t,-6-s,s,3):\ t,s\in\mathbb{R}\}$.

There are two types of variables in the 2nd example.

Lead variables & Free variables

pivot

- A variable corresponding to a **leading entry** is called a **leading variable** (or basic variable) （基变量）
- All remaining variables are called free variables （自由变量）

Remark.

A ***consistent*** *system has infinitely many solutions if and only if it has free variables.*

General solutions （通解）

Remember:

The presence of a free variable in a system does not guarantee that the system is consistent.

Example. The augmented matrix of the system

$$\begin{cases} x_1+x_2+x_3+x_4+x_5 &= 1 \\ -x_1-x_2+x_5 &= -1 \\ -2x_1-2x_2+3x_5 &= 1 \\ x_3+x_4+3x_5 &= -1 \\ x_1+x_2+2x_3+2x_4+4x_5 &= 1 \end{cases}$$

is reduced to

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right)$$

Solve the system.

No solution.

Two Fundamental Questions about a linear system: **Existence and Uniqueness** (存在性和唯一性)

1)Is the system consistent? (i.e., does a solution exist?)

2)If a solution exists, is it unique? (i.e., is there one & only one solution?

EXAMPLE: Is this system consistent?

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ -4x_1 + 5x_2 + 9x_3 &= -9 \end{aligned}$$

In the last example, this system was reduced to the echelon form:

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix} \qquad \begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ x_2 - 4x_3 &= 4 \\ x_3 &= 3 \end{aligned}$$

This is sufficient to see that the system is consistent and unique.

EXAMPLE: Is this system consistent?

$$\begin{aligned} 3x_2 - 6x_3 &= 8 \\ x_1 - 2x_2 + 3x_3 &= -1 \\ 5x_1 - 7x_2 + 9x_3 &= 0 \end{aligned} \qquad \begin{bmatrix} 0 & 3 & -6 & 8 \\ 1 & -2 & 3 & -1 \\ 5 & -7 & 9 & 0 \end{bmatrix}$$

Solution: Row operations produce:

$$\begin{bmatrix} 0 & 3 & -6 & 8 \\ 1 & -2 & 3 & -1 \\ 5 & -7 & 9 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -6 & 8 \\ 0 & 3 & -6 & 5 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -6 & 8 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

Equation notation of echelon form:

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= -1 \\ 3x_2 - 6x_3 &= 8 \\ 0x_3 &= -3 \quad \leftarrow \textit{Never true} \end{aligned}$$

The original system is inconsistent!

Two Fundamental Questions about a linear system:
Existence and Uniqueness

1)Is the system consistent? (i.e., does a solution exist?)

2)If a solution exists, is it unique? (i.e., is there one & only one solution?

Observations

- $\begin{pmatrix} 0 & 0 & \cdots & 0 & | & b \end{pmatrix}$ $(b \neq 0)$ means an inconsistent system.
- Number of lead variables = number of variables

  (i.e. no free variable)

  Each variable is uniquely determined.

  $\therefore$ Exactly one solution.
- Free variable $\Rightarrow$ infinitely many solutions

Two Fundamental Questions about a linear system:
Existence and Uniqueness

1)Is the system consistent? (i.e., does a solution exist?)

2)If a solution exists, is it unique? (i.e., is there one & only one solution?)

Theorem 2 (Existence and Uniqueness Theorem)

1. A linear system is consistent
if and only if an echelon form of the augmented matrix has no
row of the form

$$\begin{bmatrix} 0 & \dots & 0 & b \end{bmatrix}$$ (where b is nonzero).

2. If a linear system is consistent, then the solution contains either
(i) a unique solution (when there are no free variables) or
(ii) infinitely many solutions (when there is at least one free variable).

EXAMPLE: For what values of h will the following system be consistent?

$$\begin{aligned} 3x_1 &- 9x_2 = 4 \\ -2x_1 &+ 6x_2 = h \end{aligned}$$

Solution: Reduce to echelon form:

$$\begin{bmatrix} 3 & -9 & 4 \\ -2 & 6 & h \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & \frac{4}{3} \\ -2 & 6 & h \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & \frac{4}{3} \\ 0 & 0 & h+\frac{8}{3} \end{bmatrix}$$

The second equation is $0x_1 + 0x_2 = h + \frac{8}{3}$. System is consistent only if $h + \frac{8}{3} = 0$ or $h = \frac{-8}{3}$.

EXAMPLE:

$$\begin{aligned} 3x_1 &+4x_2 = -3 \\ 2x_1 &+5x_2 = 5 \\ -2x_1 &-3x_2 = 1 \end{aligned} \rightarrow \begin{bmatrix} 3 & 4 & -3 \\ 2 & 5 & 5 \\ -2 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 3 & 4 & -3 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{aligned} 3x_1 + 4x_2 &= -3 \\ x_2 &= 3 \end{aligned}$$

Consistent system, no free variables

$\Rightarrow$ **unique solution.**

EXAMPLE:

$$\begin{array}{rrrrrl} & 3x_2 & -6x_3 & +6x_4 & +4x_5 & = -5 \\ 3x_1 & -7x_2 & +8x_3 & -5x_4 & +8x_5 & = 9 \\ 3x_1 & -9x_2 & +12x_3 & -9x_4 & +6x_5 & = 15 \end{array}$$

we obtained the echelon form:

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \quad (x_5 = 4)$$

No equation of the form $0 = c$, where $c \neq 0$, so the system is consistent.

Free variables: x_3 and x_4

$\boxed{\text{Consistent system with free variables}}$ $\Rightarrow$ **infinitely many solutions.**

➢Row echelon form Algorithm

——Gaussian Elimination

Gauss, Carl Friedrich

Gaussian Elimination (高斯消元法）

- a process : $\boxed{\text{augmented matrix}} \xrightarrow{\text{row operation}} \boxed{\text{row echelon form}}$

$$\text{Row echelon form} \rightsquigarrow \text{Solutions of a system}$$

Example. Solve the system $\begin{cases} x+y = 1 \\ x-y = 3 \\ -x+2y = -4 \end{cases}$ by Gaussian Elimination.

Ans. A row echelon form is $\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right)$.

The solution is $x=2, y=-1$.

Question

Is row echelon form unique?

Ans. NO

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 0 & -3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Drawback: Row echelon form is not the simplest.

(行简化阶梯形算法)

➢Reduced row echelon form Algorithm——Gauss-Jordan Elimination

(高斯-若当消元法)

1.2 Row Reduction and Reduced Echelon Forms

(行简化和行简化阶梯形）

Reduced Row Echelon Form (行简化阶梯形)

(i) The matrix is in row echelon form.

(ii) The first nonzero entry in each nonzero row is 1.

(iii) The first nonzero entry in each row is the only nonzero entry in its column.

Example. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

Theorem 1 (Uniqueness of The Reduced Echelon Form):

Each matrix is row-equivalent to one and only one reduced echelon matrix.

How to get the reduced echelon form by the elementary row reduction algorithm?

---(After getting the echelon form by the elementary row reduction algorithm)

Step1. if the pivot is not 1, make it 1 by a scaling operation;

Step2. Beginning with the rightmost leading one, work upward and to the left to create zeros above each leading one.

EXAMPLE: Row reduce to echelon form and then to reduced echelon form:

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix} \sim \begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \sim \begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \sim \begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Gauss-Jordan Elimination

- a process : augmented matrix $\xrightarrow{\text{row operation}}$ reduced row echelon form

Final Step in Solving a Consistent Linear System: After the augmented matrix is in **reduced** echelon form and the system is written down as a set of equations:

Solve each equation for the basic variable in terms of the free variables (if any) in the equation.

EXAMPLE:

$$\begin{array}{rrrrl} x_1 & +6x_2 & & +3x_4 & & = 0 \\ & & x_3 & -8x_4 & & = 5 \\ & & & & x_5 & = 7 \end{array}$$

$$\begin{cases} x_1 = -6x_2 - 3x_4 \\ x_2 \text{ is free} \\ x_3 = 5 + 8x_4 \\ x_4 \text{ is free} \\ x_5 = 7 \end{cases} \quad \text{(general solution)}$$

Parametric description (参数形式)of solution set

The **general solution** of the system provides a parametric description of the solution set. (The free variables act as parameters.) The above system has **infinitely many solutions.**

Warning: Use only the reduced echelon form to solve a system.

Recall:

Theorem 2 (Existence and Uniqueness Theorem)

1. A linear system is consistent if and only if an echelon form of the augmented matrix has no row of the form

$$\begin{bmatrix} 0 & \ldots & 0 & b \end{bmatrix}$$ (where b is nonzero).

2. If a linear system is consistent, then the solution contains either
(i) a unique solution (when there are no free variables) or
(ii) infinitely many solutions (when there is at least one free variable).

Using Row Reduction to Solve Linear Systems

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If not, stop; otherwise go to the next step.
3. Continue row reduction to obtain the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in step 3.
5. State the solution by expressing each basic variable in terms of the free variables and declare the free variables.

Any linear system $A_{m\times n}\vec{x}=\vec{b}$: Guassian-Jordan Elimilation Method

$$\overline{A}=\begin{pmatrix} A & \vec{b} \end{pmatrix}$$

$$\overline{A}\xrightarrow{\text{Elementary Row Reduction algorithm}}\text{echelon form of } \overline{A}$$

$$\rightarrow\begin{cases} r(A)=r(\overline{A})\xrightarrow{\text{Elementary Row Reduction algorithm}}\text{reduced echelon formof } \overline{A}\rightarrow\begin{cases}\text{unique solution;}\\ \text{infinitely many solutions;}\end{cases} \\ r(A)\neq r(\overline{A})\rightarrow \text{no solution.}\end{cases}$$

Solve the following systems of linear equations.

Example 1.

$$\begin{cases} 2x_1 - x_2 - x_3 + x_4 = 2, \\ x_1 + x_2 - 2x_3 + x_4 = 4, \\ 4x_1 - 6x_2 + 2x_3 - 2x_4 = 4, \\ 3x_1 + 6x_2 - 9x_3 + 7x_4 = 9, \end{cases}$$

$$\bar{A}=\begin{pmatrix}2&-1&-1&1&2\\1&1&-2&1&4\\4&-6&2&-2&4\\3&6&-9&7&9\end{pmatrix}\to\begin{pmatrix}1&1&-2&1&4\\2&-1&-1&1&2\\2&-3&1&-1&2\\3&6&-9&7&9\end{pmatrix}\to$$

$$\begin{pmatrix}1&1&-2&1&4\\2&-1&-1&1&2\\2&-3&1&-1&2\\3&6&-9&7&9\end{pmatrix}\to\begin{pmatrix}1&1&-2&1&4\\0&-3&3&-1&-6\\0&-5&5&-3&-6\\0&3&-3&4&-3\end{pmatrix}\to\begin{pmatrix}1&1&-2&1&4\\0&-3&3&-1&-6\\0&0&0&-\frac{4}{3}&4\\0&0&0&3&-9\end{pmatrix}$$

$$\to\begin{pmatrix}1&1&-2&1&4\\0&-3&3&-1&-6\\0&0&0&1&-3\\0&0&0&0&0\end{pmatrix}\to\begin{pmatrix}1&0&-1&0&4\\0&1&-1&0&3\\0&0&0&1&-3\\0&0&0&0&0\end{pmatrix}$$

then, the general solution is

$$\begin{cases}x_1=x_3+4\\x_2=x_3+3\\x_3 \text{ is free}\\x_4=-3\end{cases}$$

Example 2.

$$\begin{cases} x_1 + 2x_2 + 5x_3 = 0, \\ x_1 + 3x_2 - 2x_3 = 1, \\ 3x_1 + 7x_2 + 10x_3 = 2. \end{cases}$$

Sol. $A = \begin{bmatrix} 1 & 2 & 5 & 0 \\ 1 & 3 & -2 & 1 \\ 3 & 7 & 10 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & -7 & 1 \\ 0 & 1 & -5 & 2 \end{bmatrix}$

$$\to \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & -7 & 1 \\ 0 & 0 & 1 & 1/2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & -5/2 \\ 0 & 1 & 0 & 9/2 \\ 0 & 0 & 1 & 1/2 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 0 & 0 & -23/2 \\ 0 & 1 & 0 & 9/2 \\ 0 & 0 & 1 & 1/2 \end{bmatrix}$$

the unique solution is

$$\begin{cases} x_1 = -23/2 \\ x_2 = 9/2 \\ x_3 = 1/2 \end{cases}$$

Example 3. $\begin{cases} x_1+x_2+x_3+x_4=1, \\ -x_1+x_2+x_3+x_4=-1, \\ -x_1-x_2+x_3+x_4=-3, \\ -x_1-x_2-x_3+x_4=-3. \end{cases}$

Sol. $\bar{A}=\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & -1 \\ -1 & -1 & 1 & 1 & -3 \\ -1 & -1 & -1 & 1 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & 2 & 2 & -2 \\ 0 & 0 & 0 & 2 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$

$\to \begin{bmatrix} 1 & 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$

$\therefore$ the system of equations has the unique solution:

$$\begin{cases} x_1 = 1 \\ x_2 = 1 \\ x_3 = 0 \\ x_4 = -1 \end{cases}$$

Exercise 1. $\begin{cases} x_1 - 2x_2 + 3x_3 - x_4 + 2x_5 = 2, \\ 3x_1 - x_2 + 5x_3 - 3x_4 - x_5 = 6, \\ 2x_1 + x_2 + 2x_3 - 2x_4 - 3x_5 = 8. \end{cases}$

Sol. $\overline{A} = \begin{bmatrix} 1 & -2 & 3 & -1 & 2 & 2 \\ 3 & -1 & 5 & -3 & -1 & 6 \\ 2 & 1 & 2 & -2 & -3 & 8 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 3 & -1 & 2 & 2 \\ 0 & 5 & -4 & 0 & -7 & 0 \\ 0 & 5 & -4 & 0 & -7 & 4 \end{bmatrix}$

$\to \begin{bmatrix} 1 & -2 & 3 & -1 & 2 & 2 \\ 0 & 1 & \frac{-4}{5} & 0 & \frac{-7}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

$\because 0 = 1$, impossible,

$\therefore$ no solution.

Exercise 2. $$\begin{cases} x_1 + 2x_2 - x_3 + x_4 = 2, \\ 2x_1 + 5x_2 - x_3 + 2x_4 = 3, \\ -x_1 - 2x_2 + 2x_3 - 2x_4 = 3. \end{cases}$$

sd. $$\bar{A} = \begin{bmatrix} 1 & 2 & -1 & 1 & 2 \\ 2 & 5 & -1 & 2 & 3 \\ -1 & -2 & 2 & -2 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & 1 & 2 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 & 5 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 2 & 0 & 0 & 7 \\ 0 & 1 & 0 & 1 & -6 \\ 0 & 0 & 1 & -1 & 5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & -2 & 19 \\ 0 & 1 & 0 & 1 & -6 \\ 0 & 0 & 1 & -1 & 5 \end{bmatrix}$$

the corresponding linear system is

$$\begin{cases} x_1 - 2x_4 = 19, \\ x_2 + x_4 = -6, \\ x_3 - x_4 = 5. \end{cases}$$

then, the general solution is

$$\begin{cases} x_1 = 2x_4 + 19 \\ x_2 = -x_4 - 6 \\ x_3 = x_4 + 5 \\ x_4 \text{ is free} \end{cases}$$

Exercise 3.

$$\begin{cases} x_1 - x_2 - x_3 + x_4 = 0 \\ x_1 - x_2 + x_3 - 3x_4 = 1 \\ x_1 - x_2 - 2x_3 + 3x_4 = -1/2 \end{cases}$$

Sol. $\bar{A} = \begin{pmatrix} 1 & -1 & -1 & 1 & 0 \\ 1 & -1 & 1 & -3 & 1 \\ 1 & -1 & -2 & 3 & -1/2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 2 & -4 & 1 \\ 0 & 0 & -1 & 2 & -1/2 \end{pmatrix}$

$$\rightarrow \begin{pmatrix} 1 & -1 & 0 & -1 & 1/2 \\ 0 & 0 & 1 & -2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

the corresponding linear system is

$$\begin{cases} \boldsymbol{x_1 = x_2 + x_4 + 1/2} \\ \boldsymbol{x_3 = 2x_4 + 1/2} \end{cases}$$

then, the general solution is

$$\begin{cases} x_1 = x_2 + x_4 + 1/2 \\ \quad x_2 \text{ is free} \\ x_3 = 2x_4 + 1/2 \\ \quad x_4 \text{ is free} \end{cases}$$

Some Terminologies

A linear system is said to be overdetermined if the number of equations is more than the number of unknowns. （超定线性方程组）

A linear system is said to be underdetermined if the number of equations is less than the number of unknowns. （欠定线性方程组）

Questions

Must an overdetermined system be inconsistent?

Must an underdetermined system have infinitely many solutions?

Is it possible for an underdetermined system to have a unique solution?

Overdetermined systems

a). $\begin{cases} x_1 + x_2 = 1 \\ x_1 - x_2 = 3 \\ -x_1 + 2x_2 = -2 \end{cases}$

No solution

b). $\begin{cases} x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - x_2 + x_3 = 2 \\ 4x_1 + 3x_2 + 3x_3 = 4 \\ 2x_1 - x_2 + 3x_3 = 5 \end{cases}$

$(0.1, -0.3, 1.5)$

c). $\begin{cases} x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - x_2 + x_3 = 2 \\ 4x_1 + 3x_2 + 3x_3 = 4 \\ 3x_1 + x_2 + 2x_3 = 3 \end{cases}$

$(1 - 0.6\alpha, -0.2\alpha, \alpha)$

Underdetermined systems

a). $\begin{cases} x_1 + 2x_2 + x_3 = 1 \\ 2x_1 + 4x_2 + 2x_3 = 3 \end{cases}$

No solution

b). $\begin{cases} x_1 + x_2 + x_3 + x_4 + x_5 = 2 \\ x_1 + x_2 + x_3 + 2x_4 + 2x_5 = 3 \\ x_1 + x_2 + x_3 + 2x_4 + 3x_5 = 2 \end{cases}$

$(1-\alpha-\beta, \alpha, \beta, 2, -1)$

➢ Rank of a Matrix (秩)

Definition

The **rank of a matrix A**, denoted by *r(A)*, is the number of nonzero rows when A is row-reduced to the echelon form.

Fact: If A is $m \times n$, then $r(A) \le m, r(A) \le n$.

Find $r(A)$ if $A=\begin{bmatrix}1 & -2 & 0 & 1\\ 0 & 1 & 1 & 1\\ 2 & -3 & 2 & 3\end{bmatrix}$.

Sol. $\begin{bmatrix}1 & -2 & 0 & 1\\ 0 & 1 & 1 & 1\\ 2 & -3 & 2 & 3\end{bmatrix}\rightarrow\begin{bmatrix}1 & -2 & 0 & 1\\ 0 & 1 & 1 & 1\\ 0 & 1 & 2 & 1\end{bmatrix}$

$\rightarrow\begin{bmatrix}1 & -2 & 0 & 1\\ 0 & 1 & 1 & 1\\ 0 & 0 & 1 & 0\end{bmatrix}$ $\quad\therefore r(A)=3.$

$$\text{Find } r(A) \text{ if } A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$\text{Sol. } \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore r(A) = 2$$

Exercises.

1. Find the rank of the following matrices.

$$(1)\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ -1 & 2 & 3 \end{bmatrix};(2)\begin{bmatrix} 1 & 2 & 3 & 4 \\ -1 & 2 & 0 & 1 \\ 0 & 1 & 0 & 2 \end{bmatrix};$$

$$(3)\begin{bmatrix} -1 & 2 & 1 & 0 \\ 1 & -2 & -1 & 0 \\ -1 & 0 & 1 & 1 \\ -2 & 0 & 2 & 2 \end{bmatrix}.$$

$$(1)A=\begin{bmatrix}1&2&0\\0&1&1\\-1&2&3\end{bmatrix}\to\begin{bmatrix}1&2&0\\0&1&1\\0&4&3\end{bmatrix}\to\begin{bmatrix}1&2&0\\0&1&1\\0&0&-1\end{bmatrix}\quad\therefore r(A)=3$$

$$(2)\begin{bmatrix}1&2&3&4\\-1&2&0&1\\0&1&0&2\end{bmatrix}\to\begin{bmatrix}1&2&3&4\\0&4&3&5\\0&1&0&2\end{bmatrix}$$

$$\to\begin{bmatrix}1&2&3&4\\0&1&0&2\\0&4&3&5\end{bmatrix}\to\begin{bmatrix}1&2&3&4\\0&1&0&2\\0&0&3&-3\end{bmatrix}\quad\therefore r(A)=3$$

$$(3)\begin{bmatrix} -1 & 2 & 1 & 0 \\ 1 & -2 & -1 & 0 \\ -1 & 0 & 1 & 1 \\ -2 & 0 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 1 \\ 0 & -4 & 0 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} -1 & 2 & 1 & 0 \\ 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -4 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 & 1 & 0 \\ 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore r(A) = 2$$

2. Suppose $r(A) = 2$, where $A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & -3 & x \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & y \end{pmatrix}$.

Find the values of x and y.

Sol. $\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & -3 & x \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & y \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 6 & 3-x \\ 0 & 0 & 0 & 0 & x \\ 0 & 0 & 0 & 0 & -x+y-2 \end{pmatrix}$

$$\Rightarrow \quad x = 0, y = 2$$

➢Application of the rank of a matrix

---How to use the rank to determine the solution type of any mxn linear system?

The solution types of any linear system ：

$$\begin{cases} \text{no solution} \\ \text{unique solution} \\ \text{infinitely many solutions} \end{cases}$$

Recall: 1.2

Using Row Reduction to Solve Linear Systems

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If not, stop; otherwise go to the next step.
3. Continue row reduction to obtain the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in step 3.
5. State the solution by expressing each basic variable in terms of the free variables and declare the free variables.

Theorem 2 (Existence and Uniqueness Theorem)

1. A linear system is consistent if and only if an echelon form of the augmented matrix has no row of the form

$$\begin{bmatrix} 0 & \dots & 0 & b \end{bmatrix} \text{ (where } b \text{ is nonzero).}$$

2. If a linear system is consistent, then the solution contains either
(i) a unique solution (when there are no free variables) or
(ii) infinitely many solutions (when there is at least one free variable).

$$\bar{A} = \left(\begin{array}{ccc:c} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

$$A$$

$$\left.\begin{array}{l} r(\bar{A}) = 3 \\ r(A) = 2 \end{array}\right\} r(A) \neq r(\bar{A})$$

How to use the rank to determine whether a linear system is consistent?

A linear system in n unknowns is consistent if and only if the rank of the augmented matrix equals to the rank of the coefficient matrix.

Example 1.

$$\begin{cases} x_1 + 2x_2 + 5x_3 = 0, \\ x_1 + 3x_2 - 2x_3 = 1, \\ 3x_1 + 7x_2 + 10x_3 = 2. \end{cases}$$

Sol. $\bar{A} = \begin{bmatrix} 1 & 2 & 5 & 0 \\ \cdots \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & -7 & 1 \\ \cdots & & & 2 \end{bmatrix}$

This system has **3** unknowns;
the rank is **3**;
the solution is unique.

$$\to \begin{bmatrix} 1 & 0 & 0 & -23/2 \\ 0 & 1 & 0 & 9/2 \\ 0 & 0 & 1 & 1/2 \end{bmatrix}$$

$r(A) = r(\bar{A}) = 3$

$\therefore$ this system is consistent.

and we can get

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -23/2 \\ 9/2 \\ 1/2 \end{bmatrix}.$$

Example 2. $$\begin{cases} x_1+2x_2-x_3+x_4=2, \\ 2x_1+5x_2-x_3+2x_4=3, \\ -x_1-2x_2+2x_3-2x_4=3. \end{cases}$$

$$\text{sd}.\ \bar{A}=\begin{bmatrix} 1 & 2 & -1 & 1 & 2 \\ 2 & 5 & -1 & 2 & 3 \\ -1 & -2 & 2 & -2 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & 1 & 2 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 & 5 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 2 & 0 & 0 & 7 \\ 0 & 1 & 0 & 1 & -6 \\ 0 & 0 & 1 & -1 & 5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & -2 & 19 \\ 0 & 1 & 0 & 1 & -6 \\ 0 & 0 & 1 & -1 & 5 \end{bmatrix}$$

$\because r(A) = r(\bar{A}) = 3 \quad \therefore$ this system is consistent.

and then we may back substitute to get

the corresponding equations : $\begin{cases} x_1 - 2x_4 = 19, \\ x_2 + x_4 = -6, \\ x_3 - x_4 = 5. \end{cases}$

we get : $\begin{cases} x_1 = 2x_4 + 19 \\ x_2 = -x_4 - 6 \\ x_3 = x_4 + 5 \\ x_4 \text{ is free} \end{cases}$.

This system has **4** unknowns;
the rank is **3**;
the solutions contains
1(=4-3) free variable.

How to use the rank to determine the solution type of a consistent linear system?

If a linear system in n unknowns is consistent, and let the rank of the augmented matrix be r, then

(i) if $r=n$, then the solution is ***unique***;

(ii) if $r<n$, then there are infinitely many solutions, and the solutions contains $n-r$ free variables.

Example 3. $$\begin{cases} x_1 - 2x_2 + 3x_3 - x_4 = 2, \\ 2x_1 - 3x_2 - 2x_3 + x_4 = 3, \\ 3x_1 - 5x_2 + x_3 \qquad\quad = 5, \\ -x_1 + x_2 + 5x_3 - 2x_4 = -1. \end{cases}$$

Sol. $$\bar{A} = \begin{bmatrix} 1 & -2 & 3 & -1 & 2 \\ 2 & -3 & -2 & 1 & 3 \\ 3 & -5 & 1 & 0 & 5 \\ -1 & 1 & 5 & -2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 3 & -1 & 2 \\ 0 & 1 & -8 & 3 & -1 \\ 0 & 1 & -8 & 3 & -1 \\ 0 & -1 & 8 & -3 & 1 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & -2 & 3 & -1 & 2 \\ 0 & 1 & -8 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -13 & 5 & 0 \\ 0 & 1 & -8 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$r(A) = r(\overline{A}) = 2 < 4, \therefore$ this system has infinitely many solutions, which contains $n - r = 4 - 2 = 2$ free variables.

and then we get $\begin{cases} x_1 = 13x_3 - 5x_4 \\ x_2 = 8x_3 - 3x_4 - 1 \\ x_3 \text{ is free} \\ x_4 \text{ is free} \end{cases}.$

Exercises

$$1.\begin{cases} x_1 - 2x_2 + 3x_3 - x_4 = 1, \\ 3x_1 - x_2 + 5x_3 - 3x_4 = 2, \\ 2x_1 + x_2 + 2x_3 - 2x_4 = 3. \end{cases}$$

$$\text{Sol. } \overline{A} = \begin{pmatrix} 1 & -2 & 3 & -1 & 1 \\ 3 & -1 & 5 & -3 & 2 \\ 2 & 1 & 2 & -2 & 3 \end{pmatrix} \to \begin{pmatrix} 1 & -2 & 3 & -1 & 1 \\ 0 & 5 & -4 & 0 & -1 \\ 0 & 5 & -4 & 0 & 1 \end{pmatrix}$$

$$\to \begin{pmatrix} 1 & -2 & 3 & -1 & 1 \\ 0 & 5 & -4 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

$$\because r(A) = 2 \neq r(\overline{A}) = 3,$$

$$\therefore \text{ no solution.}$$

$$2.\begin{cases} x_1 - x_2 - x_3 + x_4 = 0 \\ x_1 - x_2 + x_3 - 3x_4 = 1 \\ x_1 - x_2 - 2x_3 + 3x_4 = -1/2 \end{cases}.$$

$$\text{Sol. } \overline{A} = \begin{pmatrix} 1 & -1 & -1 & 1 & 0 \\ 1 & -1 & 1 & -3 & 1 \\ 1 & -1 & -2 & 3 & -1/2 \end{pmatrix} \to \begin{pmatrix} 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 2 & -4 & 1 \\ 0 & 0 & -1 & 2 & -1/2 \end{pmatrix}$$

$$\to \begin{pmatrix} 1 & -1 & 0 & -1 & 1/2 \\ 0 & 0 & 1 & -2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$\because r(A) = r(\overline{A}) = 2,$

$\therefore$ this system has infinitely many solutions, which contains $4 - 2 = 2$ free variables.

$$\text{we have } \begin{cases} x_1 = x_2 + x_4 + \dfrac{1}{2} \\ x_2 \text{ is free} \\ x_3 = 2x_4 + \dfrac{1}{2} \\ x_4 \text{ is free} \end{cases}.$$

Summary

1.An linear system is consistent $\Leftrightarrow r(A)=r(\bar{A})=r$

if $r=n$, then the solution is unique;

if $r<n$, then the system has infinitely many solutions and they contains $n-r$ free variables.

2.An linear system is inconsistent $\Leftrightarrow r(A)\neq r(\bar{A})$

where A and $\bar{A}$ denote the coefficient matrix and augmented matrix, respectively.

◆Review

Gauss-Jordan Elimination

◆Preview

Vector Equations and Matrix Equations

Glossary

1.	The system of linear equations	线性方程组
2.	Solution set	解集
3.	Matrix	矩阵
4.	(In)Consistent	(不) 相容的
5.	Elementary row operations	初等行变换
6.	Existence and Uniqueness	存在性和唯一性
7.	Echelon form	阶梯形
8.	Reduced row echelon form	行简化阶梯形
9.	Elimination	消元
10.	Gaussian elimination algorithm	高斯消元法
11.	Parametric description	参数形式
12.	Rank	秩